

FLUPSY Manager Manual

Complete operational guide for operators

Generated on: 19/05/2026

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1.1 What FLUPSY Manager is

FLUPSY Manager is the management system for shellfish (clams, oysters, mussels) pre-fattening farms in FLUPSY units (Floating UPweller SYstem). It tracks in real time baskets, life cycles, operations, seed lots, sales and transport documents (DDT).

The system combines daily operational data, growth analysis, artificial intelligence (GPT-4o) for forecasting and diagnostics, and integration with external invoicing (Fatture in Cloud) and environmental systems (Copernicus, ARPAV, ARPAE).

1.2 Key concepts

FLUPSY	Floating unit hosting the baskets. Each FLUPSY has a name, GPS location, max basket capacity.
Basket	Physical container identified by a sequential number within the FLUPSY. Can contain only one active cycle at a time.
Cycle	Life period of animals in a basket, from first activation to sale/closure. Holds all operations and a unique `cycleCode`.
Lot	Seed batch with origin, supplier, arrival date and price. Multiple cycles can come from the same lot.
Operation	Event recorded on the basket: first-activation, measure, weight, screening, sale, closure, mortality.
Size	Commercial size class (TP-180 !' TP-10000) defined by animals-per-kg ranges. More animals/kg = smaller animal.
SGR	Specific Growth Rate, daily growth percentage by weight, used for projections.
Screening (Vagliatura)	Sorting operation that redistributes animals between sieves/microns and creates new child baskets.
DDT (Transport Doc)	Transport document automatically generated from sales, with state (draft, issued, integrated).

1.3 Navigation and menu

The side menu is organized by macro-areas: Operations, Monitoring, Inventory, Analysis, Planning, Sales, System. Click on a section to expand its items.

TIP

On the Dashboard, from the top selector, you can save your preferred FLUPSY IDs: dropdowns in other pages will be filtered accordingly.

2.1 Widgets and KPIs

The Dashboard shows a real-time overview of the farm: active baskets, total animals, size distribution, marine weather, notifications and — through the IMM widget — the inventory ripeness index.

- Active baskets: number of open cycles.
- Total animals: sum of the last `animal\_count` per cycle.
- Size distribution: pie chart with animal count by size.
- Notifications: target sizes reached, deadlines, anomalies.
- Global IMM: 0-100 score, quick link to dedicated page.

FLUPSY Units and Baskets

3.1 FLUPSY management

From `FLUPSY Units` you can create, edit and delete units. Each FLUPSY has: name, description, capacity (max baskets), optional GPS coordinates, layout (rows and columns) for the visual map.

WARNING

Deleting a FLUPSY is only allowed if it has no baskets with active cycles. Use the selective deletion procedure for test installations.

3.2 Basket management

Each basket has: physical number (unique within FLUPSY), position (row/col or coordinates), state (`available`, `active`, `ready`), current cycle. State is governed by an atomic DB-level invariant: `state`, `currentCycleId` and `cycleCode` are always written together.

TIP

From `Position Management` you can move baskets on the FLUPSY map while keeping the active cycle.

3.3 NFC Tags

The NFC module allows programming physical labels to apply to baskets. The basket state can be manually overridden via NFC with timestamp and operator.

Operations

Operations are the core of the system. Every event on a basket (first activation, measure, weight, screening, sale, closure, mortality) is recorded as an operation with date, values and operator. All operations go through the `OperationsLifecycleService` to guarantee state consistency.

4.1 Operation types

first-activation	Start of a new cycle: defines initial seed (lot, animal_count, animals_per_kg, total_weight).
first-activation-from-screening	Cycle start as destination of a screening.
measure	Measurement of a small sample to estimate animals/kg.
weight	Total basket weighing to recalculate biomass.
screening	Sieve-based selection and redistribution.
closure-sale	Final sale, closes the cycle.
closure-screening	Parent cycle closure after screening.
mortality	Detected mortality recording.

4.2 Spreadsheet Operations module

Mobile-first spreadsheet-like interface: one row per basket, editable cells for fast input. Live validation, auto-save, batch on multi-selection, color-coded performance indicators.

TIP

To enter a series of quick measures across many baskets, use the Spreadsheet: the system automatically applies the "Flying shells" rule (mortality masked by initial count).

4.3 Validations and rules

- Two-layer animal count validation (frontend + backend).
- Size anomaly: regression (animals/kg increases) or jump > 1 class !' asks for confirmation.
- `measure` operations can have their date modified, respecting chronological order.
- Operation deletion always goes through `OperationsLifecycleService` to avoid state misalignment.

4.4 "Flying shells" rule

When a `measure` records mortality % "d the rolling-max mortality % of previous cycle operations, the animal count is NOT reduced. Dead shells are presumed to have already been counted at first activation and simply washed away. Only excess mortality beyond the historical max reduces the count.

NOTE

Implemented in `server/utls/misura-mortality.ts`. Admin endpoint `POST /api/admin/recalc-misure/apply` for historical recompute.

Cycles and Lots

5.1 Production cycles

Each cycle is uniquely identified by `cycleCode`. States: `active`, `closed`. At creation it's auto-classified with `qualityClass` (PREMIUM, NORMAL, SUB) based on the lot's screening history.

Cycle Report

From the cycle page you can see the full operations timeline, future growth projection and cycle-specific IMM.

5.2 Seed lots

Lots represent seed batches from external sources. Main fields: `supplier`, `arrival date`, `initial animal_count`, `animals_per_kg`, `price`. `Lot Report` shows current balance, distribution among cycles and timeline.

Screening (Selection)

6.1 Screening flow

1. Select source baskets (one or more).
2. Specify the sieves used (upper + and lower - mesh in microns).
3. Define destination baskets (also on different FLUPSY !' cross-FLUPSY).
4. Enter animal counts for each destination.
5. Confirm: the system closes source cycles and opens new destination cycles.

TIP

The `DraggableCalculator` helps distribute animals across destination baskets while keeping the total constant.

6.2 Cross-FLUPSY screening

Destination baskets can be on different FLUPSYs than sources. The screening is marked `is\_cross\_flupsy = true` and enriched with transport metadata (vehicle, operator, distance).

7.1 Sales flow

1. Create a sale selecting customer and baskets/cycles to sell.
2. Specify quantity, size and price (optionally from price list).
3. Multi-customer mode: a single source operation can generate multiple sales (one per customer).
4. Generate DDT: state `draft`, then `issued` when confirmed.
5. Fatture in Cloud and external FCloud integration with bidirectional sync.

7.2 DDT (Transport Document)

DDT contains an immutable snapshot of customer data, subtotals by size, full traceability of source cycles/lots. Three states: `draft`, `issued`, `integrated`.

WARNING

Once the DDT is integrated with Fatture in Cloud, customer data cannot be modified anymore: recreate it to correct.

Inventory and Stock

8.1 Current stock

The `Inventory` page shows real-time stock: animals by size, FLUPSY, lot. Filters by cycle state, date range and size.

8.2 Range calculation

`Range Stock` calculates stock between two arbitrary dates, reconstructing operations in the interval.

IMM — Inventory Ripeness Index

The IMM is a 0-100 score measuring how close the inventory is to the target commercial size, considering size, time, quality and data reliability. Higher = more 'ripe' and sale-ready.

9.1 Formula and components

The IMM is the weighted average of 4 components, each in the 0-100 range, weighted by the cycle's animal count. Weights are configurable and normalized to 100%.

Size (default weight 40)	How close the current size (animals/kg) is to the target. 0 = very far (animals too small), 100 = at target or beyond.
Time (default weight 35)	Days remaining to reach target according to cycle SGR. 100 = already ready, 0 = beyond horizon (default 180 days).
Quality (default weight 15)	Cycle quality class: PREMIUM=100, NORMAL=70, SUB=40, unknown=50.
Reliability (default weight 10)	Penalty on cumulative mortality: 100 at 0% mortality, 0 at `maxMortalityPct` (default 30%).

Mathematical formula

```
IMM_cycle = (wS·S + wT·T + wQ·Q + wR·R) / (wS + wT + wQ + wR)
IMM_global = :2,, "00ö7-6ÆR r æ-Ö Ç5ö7-6ÆR' ò £(animals_cycle)
```

9.2 Page KPIs

Inventory IMM	Global 0-100 score weighted by animal count.
Active cycles included	Cycles with at least one valid `measure` or `weight` (cancelled excluded).
Total animals	Sum of last `animal_count` per active cycle (compact display).
Current value (€)	Animals × €/animal price of current size (from `sales_price_list`).
Target value (€)	Animals × €/animal price at target size (potential if all ripen).
Ripe value (€)	Target value × IMM/100, realistic estimate considering current state.

9.3 Distribution and tables

Below the KPIs you'll find 4 views:

- IMM components (weighted average): Size, Time, Quality, Reliability bars to understand the main limiting factor.
- IMM bucket distribution: 0-25 (far), 25-50 (growing), 50-75 (approaching), 75-100 (ready).
- Per-FLUPSY table: compare IMM across installations to spot the most ripe ones.
- Per-Lot table: see which lots are ripening faster.

9.4 Trend and daily snapshots

Every night at 03:30 the scheduler saves a snapshot to `imm\_snapshots` (transactional to avoid data loss). The history chart shows the last 90 days with global IMM and all 4 components.

TIP

Use the `Snapshot now` button to force a manual save, useful after big screenings.

9.5 Order coverage

The `Order coverage by size` table compares:

- Demand: animals required by orders (parsing `name` field with regex `screen(\d+)`).
- Supply: animals available at that size (sum of active cycles).
- Ready (IMM"e75): only those actually ripe.
- Gap: demand - ripe_supply. Negative = surplus.
- Coverage %: how much demand is covered by ripe supply.

9.6 Persistent configuration

From the `Weights and thresholds` card you can edit and save to DB: the 4 weights, horizon (days), fallback SGR, baseline/max mortality, default target size.

NOTE

Saved parameters are used as base; URL query parameters temporarily override them.

9.7 Excel export

The `Export Excel` button generates a file with 6 sheets: Summary, Distribution, By FLUPSY, By Lot, Active Cycles, Order Coverage. Useful for offline analysis or sharing with corporate ERP.

9.8 Interpretation and operational use

0-25	Inventory far from target: small animals or very young lots. Plan long growth cycles.
25-50	Growing. Monitor SGR and mortality, evaluate selection screenings.
50-75	Approaching target: start sales planning.
75-100	Ready for sale. Check orders and fulfillment priorities.

TIP

Compare Size and Time components: if Size is high but Time is low, animals are large but close to target. If Time is high but Size is low, there's an inconsistency — check the cycle SGR.

WARNING

IMM does not automatically include order deadlines because shared orders lack `due\_date`. Add delivery dates to enable urgency boost.

Environmental Log

10.1 Environmental data

On each login the system automatically captures a snapshot of marine data from Copernicus/Open-Meteo network (temperature, salinity, oxygen, current) and from ARPAV Vallona and ARPAE Gorino 2 buoys. The dedicated page shows trend charts, table and CSV export.

11.1 AI features (GPT-4o)

- AI Dashboard: KPIs with AI-generated commentary.
- Recommended Activities: daily prioritized list generated by analyzing basket state.
- AI Report Generator: PDF reports with narrative.
- AI Enhanced (beta): natural language queries on the database.
- Variance Analysis: AI comparison of budget, orders and projected production.
- Anomaly detection on measure operations.

12.1 Growth projection

Day-by-day simulation of current inventory growth until target size, with mortality incorporated. Shows monthly progression and when each cycle will be ready.

12.2 Optimized Sales Planning

Dual-engine module (greedy + LP) balancing cash, revenue and order coverage over 6-36 month horizons. Three modes: `cash` (maximize liquidity), `balanced`, `revenue` (maximize turnover).

System, Backup and Security

13.1 Automatic backup

Daily automatic database backup with 7-day retention. Manual restore from `Database Backup` page. Audit logging for all critical operations in `audit\_logs`.

13.2 Nightly integrity check

Automatic scheduler verifying basket state consistency (state/currentCycleId/cycleCode), orphan records and FK violations. Any anomaly is notified.

13.3 Environment variables

- `DATABASE\_URL`: PostgreSQL connection.
- `OPENAI\_API\_KEY`: OpenAI key for AI features.
- `FATTUREINCLOUD\_\*`: invoicing integration tokens.
- `EXTERNAL\_API\_KEY`: external endpoint protection.

14.1 Common issues

Inconsistent basket state	Run nightly integrity check or use admin re-alignment endpoint.
IMM = 0 but cycles exist	Cycles may lack a valid <code>`measure`/`weight`</code> . Enter a measure.
DDT not integrated	Check Fatture in Cloud token in settings.
WebSocket disconnected	Refresh the page; in dev it's normal during hot-reload.
Wrong target-size notification	Check units: <code>`animals_per_kg`</code> is read directly, not computed from weight.

Appendix: Size Scale

15.1 Scale rules

WARNING

The ID in the `sizes` table is ARBITRARY: never compare sizes using ID.

The only correct way is to check which `(min\_animals\_per\_kg, max\_animals\_per\_kg)` range the measured `animals\_per\_kg` falls into, then sort by `min\_animals\_per\_kg` ascending.

- More animals/kg != smaller animal != smaller commercial size.
- Fewer animals/kg != bigger animal != bigger commercial size.
- TP-10000 (801-1,200/kg) = biggest. TP-180 (70M-100M/kg) = larvae.

15.2 Weight units

Database	All weights in GRAMS.
Input forms	Accept GRAMS with clear label.
Display "Weight (kg)"	Values are divided by 1000 in the UI.

